Data Structures and Amortized Analysis — Exercises

A04-E01 Transfer only when T is empty

Starting empty, execute ENQ a, ENQ b, DEQ, ENQ c, DEQ, DEQ. List both stacks bottom to top. Record each state, returned item, primitive cost, and $\Phi = 2|S|$. Which dequeue fails if S is transferred even when T is nonempty?

A04-E02 How large must the potential coefficient be?

Let $\Phi_a = a|S|$ for a constant $a \geq 0$. Derive charges for ENQUEUE, an ordinary DEQUEUE, and a DEQUEUE transferring k records. Which a give all three a constant upper bound independent of k?

A04-E03 A nonempty initial state is not free

Initially S=[a,b,c,d] bottom to top and T is empty. Execute one DEQUEUE. Compute c, the potential change, and the charge for $\Phi = 2|S|$, and verify the telescoping identity. Why does charge 1 not mean actual cost 1?

A04-E04 A complete growth-and-contraction ledger

Use the stated APPEND / DELETE-LAST contract. Starting at (0,0), append a,b,c,d,e and delete e,d,c,b,a. List (n,M), records moved, actual cost, potential, and charge after each operation. Verify the totals.

A04-E05 A counterexample to shrinking at half occupancy

Under the bad half-full shrinking rule, append eight records from empty, then repeat APPEND x / DELETE-LAST four times. Give the state cycle and exact costs, compare with 3m, and generalize to a family with superlinear total cost.

A04-E06 Does adding a constant to potential break the proof?

Suppose $\Phi \geq 0$ and the empty initial state has $\Phi_0 = 0$. Replace it with $\Psi = \Phi + 7$. Is the argument invalid because $\Psi_0 \neq 0$? Derive the correct total bound.

A04-E07 Constant amortized cost is not constant latency

Both the two-stack queue and dynamic array have constant amortized bounds. Give a reachable state causing one linear-cost call for each, and explain what is missing if every call must meet a fixed latency budget.

A04-E08 Middle deletion is outside the original theorem

Add DELETE-FIRST to the contiguous array, preserving order by shifting every remaining record. Append n records, then perform n DELETE-FIRST operations. Even without contraction, is total work O(n)? Give the move sum and identify the changed contract.

A04-E09 A four-bit wraparound ledger

A four-bit counter increments modulo 16, initially at 1101. Perform three increments. List the bit string, flips, potential, and amortized charge after each call. What assumption is missing from a claim that actual cost is at most 2×3?

A04-E10 Charging units for MULTIPOP and MULTIPUSH

Starting empty, execute PUSH a, PUSH b, PUSH c, MULTIPOP(2), MULTIPOP(9). Count successful pushes/pops only. Give actual cost and bank balance, charging PUSH 2 and other calls 0. If one MULTIPUSH(k) call inserts k items, can each API call still be charged a constant?

A04-E11 An explicit increment/decrement counterexample

Start a four-bit counter at zero. Increment seven times, then perform four INCREMENT/DECREMENT pairs. Give preparation cost, alternating states, cost per pair, and total cost. Generalize to k bits while including preparation calls, and disprove constant amortized cost.

A04-E12 Does a rejected enqueue preserve FIFO?

Use capacity C=3. Starting empty, enqueue A,B,C,D, dequeue once, enqueue E, then dequeue three times. Give h, s, physical storage, and returns after each call. May the failed insertion of D advance h?

A04-E13 Why reversal saves nxt first

Reverse a→b→c→NIL. After each iteration, list paths from prev and cur and explain the invariant. What happens if cur.next=prev is followed directly by cur=cur.next?

A04-E14 A sentinel list with one data item

Initially z.next=z.prev=z. Insert x after z, then delete x. Give the targets and values of all four insertion writes and both deletion writes. Why is deleting z invalid?

A04-E15 Does a recycled handle still identify the old object?

Use the five-slot pool in the text. Allocate slot 1, append D after C, then remove and free B at slot 2. Give head, free, and both chains. Which slot is allocated next? What is wrong with treating an old handle 2 as B?

A04-E16 Wraparound, deletion, and updating a duplicate

Use m=5 and linear probing with home(k)=k mod5. Execute INSERT(4,a), INSERT(9,b), INSERT(14,c), DELETE(9), INSERT(14,c′), INSERT(19,d), FIND(24). Give probe paths, final records, and live count. Why not insert 14 immediately at DELETED?

A04-E17 Enumerate one affine family

Let P=5,m=3, uniform a∈{1,2,3,4} and b∈{0,1,2,3,4}. For x=1,y=4, list colliding b values for each a and compute the probability. Must it equal 1/3 to be universal?

A04-E18 The cost of a noncoprime step

For m=8,home=1, list probe orders and periods for steps 2 and 3. Can step 2 insert when all reachable slots are full but slot 0 is empty? List triangular offsets i(i+1)/2 mod8 and specify when division occurs.

A04-E19 An exact expectation versus its upper bound

Let m=4, slots 0,1 occupied and 2,3 EMPTY. A query has a uniform permutation among all 24. Compute Pr(X≥1), Pr(X≥2), Pr(X≥3), and E[X]. Compare with 1/(1−α). Why is the probability that the first two slots are occupied not α^2?

A04-E20 FKS space and membership

Use the six-key FKS layout. Compute secondary slots S and first-level collision pairs Y and verify S=n+2Y. Trace queries 26,14,5,10. Which results fail if any occupied slot is treated as success?

A04-E21 Why deleting everything need not make search cheap

Fill a linear-probe table of size 5 with keys 0–4, then delete all keys, leaving tombstones. Compute α,β and FIND(7)'s path. Does this refute the unsuccessful-search theorem for ideal uniform probing?

A04-E22 Which step fails for a composite modulus?

Change the one-dimensional MIT dot-product family to modulus 4: uniform $a \in \{0,1,2,3\}$, $h_a(x)=ax \bmod 4$. For x=0,y=2, compute collision probability and identify the failed step of the prime-modulus proof.

A04-E23 Local comparisons can miss an ancestor violation

Let root=20, with children 10 and 30, and 10.right=25. Every parent-child comparison seems valid. Which ancestor interval fails? Trace FIND(25) on these links.

A04-E24 Same tree, different query distributions

Use root 5 with children 2,8, plus 2.right=3 and 8.left=7. Compute mean comparisons for uniform successful queries, uniform integers 1..10, and a distribution concentrated entirely on query 9.

A04-E25 Two routes to the successor

On the preceding tree, trace SUCCESSOR(5),SUCCESSOR(3),SUCCESSOR(8), including upward and downward steps. Why does absence of a right child not imply absence of a successor?

A04-E26 An immediate successor must skip the deeper-successor branch

Let root z=20, z.left=10,z.right=y=40,y.right=50. What happens if y.right=z.right is executed without checking y.parent!=z? Give the correct final links.

A04-E27 Root deletion when the successor has a right child

Insert 20,10,40,30,25,27,50. Delete 20 by transplanting and list all changed child/parent links. How would copying the successor payload affect external handles differently?

A04-E28 Expected height over six insertion orders

Enumerate all six insertion orders on {1,2,3}. With edge height, compute $E[X_3]$,$E[2^{X_3}]$, compare $2^{E[X_3]}$ with $E[2^{X_3}]$, and compute the mean height under uniform sampling of the five shapes.

A04-E29 Fixed-rank versus uniform-rank expected search cost

Build a BST from a uniform order on five distinct keys. Compute expected depth and successful comparisons for rank 3, then expected mean comparisons for uniform queries among the five keys. Show harmonic values and substitutions.

A04-E30 Bases and inequalities in the exponential-height proof

A proof writes both $Y_0=2^{-1}$ and $Y_0=0$, then uses E[max(U,V)]=max(E[U],E[V]). Identify the errors. Give a valid recurrence and finish the bound from $a_n \le n^3$ to $E[X_n] \le 3\log_2 n$.

A04-E31 • Black-height versus full-path black count

Root 20B has children 10R,30R and black NIL at all other child positions. Find bh(20), bh(10), bh(NIL), and the full-path black count including the root. Recompute after blackening both children.

A04-E32 • Outside parent and an empty middle subtree

Given p30.left=x10, x.right=y20, y.left=β15, list the six writes for a left rotation. Which two change for a root rotation? Which is omitted for β=NIL? Why is “make y's old parent point to x” wrong?

A04-E33 • Black counts for non-root I1

Let g be black and p,u red. Each path below p/u, excluding p/u and including terminal NIL, has b blacks. I1 makes g red and p,u black. Compute bh(g) and the full-path count through g before and after. What changes when g is the root and is finally blackened?

A04-E34 • I2 cannot return early

Insert 35,15,25 into an empty tree. At insertion, after I2, and after I3, list root, parent/child links, colors, and the record referred to by z. Check the mirrored order 15,35,25 too.

A04-E35 • Counting boundaries in the height proof

Use internal edge height h and root-to-external-NIL edge height H. Prove the subtree lower bound $2^{bh(x)}-1$ and derive $H \le 2\log_2(n+1)$. Give h,H for empty and singleton trees. Why is O(log(n+1)) not a uniform operation bound including the empty case?

A04-E36 • Compare repair schedules precisely

Critique: “Each level needs at most two rotations, so CLRS insertion performs O(log n) rotations; top-down insertion is merely a different translation.” Distinguish time, primitive rotation count, and algorithm version.

A04-E37 • Original versus relocated color

Insert 40,20,60,10,30,50,70,45,55, then delete 40. Identify z,y,x and the original/final colors of y. Track NIL.parent. Why must repair not be selected from the final color of y?

A04-E38 • Last-node deletion and red-child absorption

Delete 25 from a singleton 25B, and separately from root 25B with its only child 15R. Does the loop execute? Give the final root/color and explain why extra black is not a third stored color.

A04-E39 • Why D1 must continue into D2

Insert 15,25,35,45,55,65, then delete 15. Track x,p,w through D1 and D2, including the helper return value. Give all final colors and identify the node that absorbs extra black.

A04-E40 • Path counts through D3 and D4

Insert 15,25,45,35, then delete 15. Starting from paths 25B→NIL(x) and 25B→45B→35R→NIL, track actual blacks, extra black, and effective totals before D3, after D3, and after D4. Give the final tree.

A04-E41 • Right deficit and an immediate successor

Insert 15,35,45,25, then delete 35. Identify the successor and the required parent of x=NIL. Trace mirrored D3/D4 by near/far roles and give the final tree.

A04-E42 • Why deletion has at most three rotations

Prove that only D2 propagates repair upward, and that D1,D3,D4 each occur at most once per deletion. Why does at most three rotations not imply O(1) deletion?

A04-E43 • Local and global ranks

Using the ten-key tree in the text, trace select(7), then compute rank for node 27. Show every k, local r, and added prefix. State contracts for empty trees and duplicate keys.

A04-E44 • Summaries after successor relocation

Delete 20 from the same ten-key tree. Its successor is 25 with 25.right=27. Give the size-repair start, order, and values. Why is starting at the old root's parent wrong?

A04-E45 • A local formula does not imply two-node-only repair

Why do size and max need only local rotation repair while height may affect ancestors? State the left-rotation update order and the cost of storing every node's global rank explicitly.

A04-E46 • A larger max is not necessarily safe

Change [8,15] to [8,27], then delete it, incorrectly leaving [5,7].max=27. Trace query [22,22], identify the missed answer, and use the general pruning proof to explain why an upper bound is insufficient.

A04-E47 • Touching endpoints and field updates

Compare [4,5] and [5,7] with their half-open versions. Give both overlap and left.max tests. What must change when editing low or only high?

A04-E48 • The cost of reporting all answers

Repeatedly search and temporarily delete each match, then reinsert all matches. With k answers, why does the full bound contain k+1 rather than just k? Does this establish O(log n+k)?

A04-E49 • Capacity parameters and split timing

Let order m mean maximum children. Give nonroot key ranges for m=3, m=4, and t=2. Why can a full order-3 leaf [2,8] not be split before insertion?

A04-E50 • Height and page reads for a billion keys

For t=100 and n=10^9, use sparse and dense bounds to find possible heights. How many page reads reach a deepest leaf if only the root is resident?

A04-E51 • Every child pointer in an internal split

For t=2, parent[50] has full left internal child[10,20,30] with leaves[5],[15],[25],[35], and right child[70] with leaves[60],[80]. Split the full left child. List all parent keys and child pointers; verify each key appears exactly once.

A04-E52 • A deletion that borrows and then merges

Use the text's delete 5 tree. Track parent 20, sibling 30, and subtree 25, then give the final tree after merging[5] and[15]. Which range fails if keys move but subtree 25 does not?

A04-E53 • Payload consistency during internal replacement

Two t=2 trees have root[20]. TreeA has leaves[5,10],[30]; treeB has[5],[30,40]. Delete 20 from each, giving replacements and final trees. What if record 20's payload stays at the root?

A04-E54 • Root repair after a failed deletion

For t=2, root[20] has leaves[5],[30]. Delete 99. Give the return value, merged-child index, and final tree. What fails if the wrapper returns immediately on NOT_FOUND?

A04-E55 • Why only the root may contract

Start with root[20] and leaves[5],[30]. Delete 20, then 5, then 30. Give each root and height. Why may an arbitrary internal node not be replaced by its sole child?

A04-E56 • Which costs improve with within-page binary search?

Separate page-access and CPU bounds for search, insertion, and deletion. Why does binary-search lower_bound not make every operation O(log t) per level?

A04-E57 • Circular lists and the meld contract

Give the pointer updates that splice two nonempty circles entered at a and b. Why must the old heaps not remain independently usable after destructive UNION?

A04-E58 • Complete consolidation and child promotion

Insert 1,4,9,7,12,6,15,10 and extract 1 in the order used in the text. List every link and the final forest. Next extract 4, processing old roots 10,6 before promoted children 9,7. Give the new forest and minimum.

A04-E59 • A decrease moves a subtree, not just a key

Start each case from the chain 1→12→15→20 with unmarked 12. Decrease 15 to 14 or to 11, recording roots, parents, degrees, marks, and the minimum.

A04-E60 • Root and nonroot grandparent outcomes

In the two cascading examples, decrease x=13 to 1 with p=8 already marked. Give roots, marks, and (r,m) after both cuts for each case; do not simply say that g is marked.

A04-E61 • Constants in the potential proof

A cascade cuts c nodes and costs at most a(c+1). Prove constant amortized cost with the same Φ=K(r+2m). Why is the expression O(c)−c=O(1) alone insufficient?

A04-E62 • Bound degree without assuming logarithmic height

Why does the ith child by attachment time currently have degree at least i−2? Use strong induction on subtree size to prove size(x)$\geq F_{d+2}$, then bound the maximum degree.

A04-E63 • Duplicate minima and deletion identity

Distinct nodes u and x both have ordinary key −∞. When can DELETE(x) fail if it only assigns −∞ again? Give a comparison rule ensuring removal of x and its amortized cost.

A04-E64 • The actual cost of deferred work

Insert n distinct keys into an empty heap, then extract the minimum. Why can that extraction cost Θ(n) without violating the logarithmic amortized bound? What extra term is needed when starting from an arbitrary existing forest?

A04-E65 • Odd widths and local coordinates

At u=32, split 21 and 30. Then split 21's low part inside its cluster. List local u,H,L,high,low and reconstruct each key.

A04-E66 • Reconstruct the set and trace three successors

For S={2,3,7,18,21,30},u=32, list extrema, local sets in nonempty clusters, and summary. Trace successor queries 3,20,30.

A04-E67 • Predecessor is not a perfectly symmetric copy

Find predecessors of 3,2,31 in the same set. Which answer fails if a missing summary predecessor always returns NIL, and why?

A04-E68 • Swap out min and activate a new cluster

Starting from the original set, insert 1 then 10. List extrema, changed clusters and summary, identifying the potentially recursive summary update.

A04-E69 • Promote min and reduce to a singleton

Delete 2 separately from S={2,3,7,18,21,30} and T={2,3}. Track promotion, removed offset, summary, max, and the reassignment of x.

A04-E70 • The actual cost of two recursive calls

Prove that deleting from a cluster and then possibly from the summary cannot both incur nonconstant recursion. Derive the time bound including odd widths and the u=2 base.

A04-E71 • Space induction for full allocation

For u=HL with high/low sizes H,L, assume $P(u) \le HP(L) + P(H) + cH$. Prove a linear bound using $P(u) \le A(u-1) - B$, specify conditions on A,B, and account for initialization.

A04-E72 • What invalid update preconditions can destroy

At u=4 with set{1}, what does DELETE-PRESENT(2) do? What invariant can INSERT-ABSENT(1) break on the same singleton? Give safe public wrappers and bounds.

References and attribution

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https://github.com/xinhi6716/computer-science-self-study